Name: _____

Section: _____

Instructions: Show supporting work to receive full credit for each problem. No calculators, notes, or other external resources are allowed. The quiz will be graded out of 10 points.

1. Consider the first-order ODE

$$\frac{dy}{dx} = y^2.$$

(a) Verify that $y = -\frac{1}{x+C}$ is a 1-parameter family of solutions to the ODE, assuming an appropriate interval of existence.

(b) If we impose the initial condition $y(1) = 4$, find the unique solution to the IVP and the largest interval of existence on which it is defined.